Fiscal Year:	FY 2024	**Task Last Updated:**	FY 05/01/2025
PI Name:	Brown, James B. Ph.D.		
Project Title:	Variation in CNS Damage Signaling and Blood Sentinels of Neuropathology After Exposure to Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jbbrown@lbl.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	1 Cyclotron Rd		
PI Address 2:			
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 HERO 80JSC017N0001-Crew Health and Performance (FLAGSHIP1, OMNIBUS). Appendix A-Flagship1, Appendix B-Omnibus
Start Date:	05/01/2022	**End Date:**	01/31/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 01/31/2024 per A. Whitmire/NASA-JSC (Ed., 4/25/24).		
Key Personnel Changes/Previous PI:	NOTE: The previous Principal Investigator (PI) was Dr. Andrew Wyrobek at the Lawrence Berkeley National Laboratory (LNL) from April 2019-April 2022. Dr. Wyrobek retired ain 2022. Dr. Ben Brown, also at LNL, took over as PI in May 2022 and the project was extended.		
COI Name (Institution):			
Grant/Contract No.:	80JSC019T0007		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: New aims per the PI report; see Task Progress section for additional information (Ed., 10/13/24). Aims 1-2: Integration of independent single-ion data sets provided by NASA and determination of how strain/genotype, sex, age at exposure, diet, and time between exposure and behavioral assessment modify the effects of single-ion radiation on different operation-relevant behavior domains. Aim 3: Apply advanced statistical methods to integrate the risk estimates from Aim 1 and Aim 2 into a unified model to predict individual-level risk for differing behavioral domains. Aim 4: Investigate the associations between CNS subregion protein biomarkers and behavior performances of individual animals on anxiety and memory assays, and evaluate computational models using on CNS tissue biomarkers to predict anxiety and memory performances of individual irradiated and unirradiated animals after exposure to space-relevant ions. NOTE: This project is a continuation of the investigation titled, Variation in CNS Damage Signaling and Blood Sentinels of Neuropathology After Exposure to Space Radiation", with Dr. Ben Brown from the Lawrence Berkeley National Laboratory (LNL) as the new Principal Investigator (PI). (Ed., May 2022). Major objectives of the NASA space radiation research program are to enable human exploration of space without exceeding limits for immediate and persistent risks to the central nervous system (CNS) from space radiation. The proposed research will investigate the CNS subregions of rodents exposed to simulated space radiation for molecular indicators for vascular damage, inflammation, and neurological abnormalities after space radiation. This project will apply multi-omic technologies (proteomics, metabolomics, and bioinformatics) to archived CNS brain subregions from irradiated mouse and rat behavioral models. The specific aims are: Aim 1. Characterize the persistence of radiation-induced molecular abnormalities in cortex and hippocampus after low-dose exposures to 56Fe particles, and compare the predictions for CNS tissue damage and late-onset neuropathologies in similarly irradiated mice and rats. Aim 2. Identify persistent bio-effect markers in peripheral blood and cerebrospinal fluid (CSF) that correlate with molecular damage in CNS vascular or immune functions. Our research plan will provide testable hypotheses of CNS tissue damage and identify molecular targets for susceptible pathways/functions of CNS damage. This project will also provide proof-of-principle whether CNS damage relevant bio-effect metabolites can be detected in CSF and blood. This project will also identify radiation-sensitive pathways, suitable for future development of biological countermeasures to reduce CNS risks from space radiation. The results of this research are designed to help NASA reduce the uncertainty associated with during mission behavior and CNS risk for astronauts on deep space exploration missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This is the final technical report for the “Predictive Behavior Modeling Project”, a research project supported by supplemental funding to the original NASA Award 80JSC019T0007. Grant #80JSC019T0007, entitled "Variation in CNS Damage Signaling and Blood Sentinels of Neuropathology After Exposure to Space Radiation", was managed as two separate projects, both awarded to Lawrence Berkeley National Laboratory (LBL) (Andrew J. Wyrobek, PI). Several years after the original grant 80JSC019T0007 was funded, NASA provided supplemental funding for the Predictive Behavior Modeling Project: “Predictive Modeling of Cosmic Radiation Effects on CNS Tissue and Operation Relevant Performance”, to be managed as a sub-account to the original grant. The two projects have had different research goals and separate research teams with some overlap of personnel. Both were seriously delayed by the COVID wet laboratory shutdowns at LBL and University of California, Berkeley (UCB) in 2020-2022, which modified the research expectations, aims, timelines, and plans, as approved by NASA management. The funding for both component projects terminated on January 31, 2024. In this investigation, we analyzed the integration of independent single-ion data sets provided by NASA and evaluated how dose, radiation type, genotype, sex, age at exposure to radiation, diet, and time between exposure and behavioral assessment modify the effects of single-ion radiation on behavior. AIM 1-2 Conclusions Data from rats irradiated by Rabin et al. were analyzed using state-of-the-art statistical and modeling methods to assess dose-response relationships for behavior endpoints and associated uncertainties. Some key findings are: • Significant dose-response relationships were observed for the behavioral endpoints studied for both 56 Fe and 4 He. • Modifying factors had a significant influence on the effects of 56 Fe and 4 He on some behavioral outcomes, but not others. For instance, genotype was observed to modify the effect of 56 Fe on anxiety (EPM) and the effect of 4 He on the bar press (BP; motivation) performance. Sex was observed to modify the effect of 4 He on novel object recognition (NOR; recognition memory), as well as novel object location (NL; spatial learning and memory), but we did not have enough data to evaluate the effects of sex following 56 Fe exposure. Age at exposure was observed to modify the effect of 56 Fe on BP performance, but we did not have enough data to assess the effect of age at 10 exposure following 4 He exposure. Diet was observed to modify the effect of high doses of 56 Fe on recognition memory (NOR), but we did not have enough data available to evaluate diet following 4 He exposure. Time of behavioral assessment post-exposure was observed to modify the effects of 56 Fe on anxiety (EPM), BP performance, and recognition memory (NOR), as well as the effects of 4 He on the BP performance, recognition memory (NOR), and spatial learning and memory (NL). Behavioral responses during an extended deep space mission may be influenced by a combination of many factors in addition to radiation – sleep, stress, O2 concentrations, etc. It is also possible that humans may not exhibit effects at very low space-relevant doses detected in rodents because of the complexity of human thought processes that have a large capacity to compensate for deficits in various domains of functioning (Duca and Rose 2011). Human cognitive and behavioral data from space travel are currently very limited. Some cognitive deficits have been observed, such as impaired concentration, short-term memory loss, and inability to multitask (Krittanawong et al. 2023). Also, sleep disorders in flight are a common concern that can lead to functional and behavioral deficits. Presently, data are not available for radiation-induced central nervous system (CNS) effects in astronauts; this remains an uncertainty that

requires extrapolation from experimental animal studies. This makes the evaluation of the available rodent behavioral data critically important to help identify the most relevant gaps, and thereby focus future research on the most urgent needs for human deep-space missions. Although there have been a substantial number of rodent studies published using space-relevant ions, the doses employed have generally been too large to be relevant to those expected in space, and the extrapolation of higher dose data to lower doses may not be possible given the complex dose and Linear Energy Transfer (LET) responses observed for behavioral deficits in rats.

AIM 3 Conclusions

We found distinct differences in low-dose response among behavior endpoints. NOR and Attentional Set-shifting (ATSET) showed maximal responses between sham and 1cGy exposures (or lowest dose tested). These endpoints predominantly test hippocampus-perirhinal-cortex and prefrontal cortex functions, respectively. EPM showed near-linear responses. These endpoints test amygdala and striatum (dopamine) functions, respectively. While several ion exposures exhibited a lack of statistically significant impacts on ATSET endpoints at higher doses (10-15cGy), the overall goodness of fit test did not significantly prefer a non-monotonic model for any individual ion, nor was there a statistically significant trend towards non-monotonicity. NOR, in contrast, showed non-monotonic, U- or W-shaped dose-response curves, with higher doses returning to the state of sham exposures and exhibiting no deficits. The biology behind this intriguing difference merits further study. The collection of additional ATSET data at both lower and higher doses may reveal that the tendency of 15cGy doses to return to the state of sham is similar to NOR, but with the data available, this is unclear. The observation that response is often maximal between sham and 1cGy across multiple behaviors and ion exposures indicates that additional studies at low doses – less than 1cGy – are needed to exhaustively map the dose-response curves for behavioral deficits upon exposure to space radiation.

REFERENCES

Duca L, and Rose J. Functional Compensation. In: Kreutzer, J.S., DeLuca, J., Caplan, B. (eds) Encyclopedia of Clinical Neuropsychology. Springer, New York, NY. 2011; https:// 0-387-79948-3_400

Krittanawong C, Singh NK, Scheuring RA, Urquieta E, Bershad EM, et al. Human Health during Space Travel: State-of-the-Art Review. Cells 2023; 12, 40. https://

Bibliography Type: Description: (Last Updated: 05/06/2025)

Articles in Peer-reviewed Journals

Straume T, Mora AM, Brown JB, Bansal I, Rabin BM, Braby LA, Wyrobek AJ. "Non-DNA radiosensitive targets that initiate persistent behavioral deficits in rats exposed to space radiation." Life Sci Space Res. 2025 May:45:44-60. https://doi.org/10.1016/j.lssr.2024.12.003 ; PMID: 40280642 , May-2025